

test

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arithmetic: $\sum_{k=1}^n (a + (k-1)d)$

1. $S_n = a + (a + d) + (a + 2d) + \dots + (a + (n-1)d)$
2. $S_n = (a + (n-1)d) + (a + (n-2)d) + \dots + a$
3. $2S_n = [a + (a + (n-1)d)] + [(a + d) + (a + (n-2)d)] + \dots$
4. $2S_n = n(2a + (n-1)d)$
5. $S_n = \frac{n}{2}(2a + (n-1)d)$

geometric: $\sum_{k=1}^n ar^{k-1}$

1. $(S_n - rS_n) = (a + ar + ar^2 + \dots + ar^{n-1}) - (ar + ar^2 + ar^3 + \dots + ar^n)$
2. $S_n - rS_n = a - ar^n$
3. $S_n(1 - r) = a - ar^n$
4. $S_n = \frac{a - ar^n}{1 - r} = \frac{a(1 - r^n)}{1 - r}$
 $r \neq 1$

converging infinite partial sums + a little dE proof for fun